

CIE

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TITLE: Smart AI Cataloging Application

SCHOOL: Chitkara University Institute of Engineering and Technology

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Name: Shivansh Sood

Roll No/ Employee Code: 2210992328

ALL MEMBERS DETAILS INCLUDING SUPERVISOR

NAME	EMP CODE/ROLL	ADDRESS WITH EMAIL AND MOBILE NO.
Shivansh Sood	2210992328	UP, 7289840657
Sahil Arya	2210990757	Chandigarh,sahil757.be22 @chitkara.edu.in , 8847483297
Rishi Kumar	2210992161	Bihar, 6202216144
Dhruv Sood	2210990279	Mohali, 8295566912
Dr. Rajat Takkar	CET1004342	rajat.takkar@chitkara.edu.i n , 9467026112

ABSTRACT

1.1 Problem Background

The conventional method of handling inventories and the workflow of e-commerce businesses is plagued with problems associated with manual cataloging. This approach involves tagging and categorizing the items manually, which is susceptible to human error and inconsistent naming of items, resulting in considerable delays as the size of the inventory increases. Organizations tend to encounter data silos in the form of unstructured data such as pictures and descriptions of the items that cannot be easily filtered or searched. Consequently, the process of introducing new items into the market takes time and is costly.

1.2 Objective of the Project

The main purpose of this task is to create a "Smart Cataloging Application," which is intended to automate the cataloging process of the stock list. The use of Artificial Intelligence will help to make an analysis of the data about the product, such as pictures and text information, and put it into the appropriate category.

1.3 Proposed Solution / System Overview

The recommended system will be a complete end-to-end solution using Next.js that combines Computer Vision and NLP models for classification tasks. It will adopt a serverless architecture in which the users will upload product information, and the backend will send that data to the AI engine such as the GPT-4o from OpenAI or a Vision-based model to extract the features. The system uses a "Smart Suggestions" interface that allows users to preview the AI-generated tags before entering the main database.

1.4 Key Features or Functionality

Important functionality involves AI-Powered Image Recognition for automatic categorization and AI-Powered Attribute Extraction such as color, material, brand from unstructured texts. These include an inventory dashboard, built using Next.js for improved speed, the ability to upload images in bulk, and a search engine that uses artificial intelligence tags. Furthermore, there is also an edit-and-confirm process to ensure human in the loop for edge cases.

1.5 Technology / Method Used

In terms of technology stack, the application makes use of the MERN or T3 Stack (Next.js, Tailwind CSS, and PostgreSQL or MongoDB as the database). API routes using Next.js provide the connection between the front-end and the back-end (models used for AI inference). In the case of image analysis, OpenAI Vision API or TensorFlow.js is incorporated into the process. The methodology involves the use of Agile: starting with building the classification engine, then

Next.js front-end, and finishing with performance improvement using edge caching and database indexing.

1.6 Expected Outcome & Benefits

The anticipated result is a smart and dynamic website capable of saving up to 80% of the time spent on categorizing products. Its main advantages will be unprecedented accuracy in assigning tags to the inventory, a significant decrease in expenses, and improved convenience for users in searching. As for the boring tasks in the input process, they will be done automatically, allowing companies to increase their inventory quickly without hiring additional personnel..

1.7 Original Contribution Statement

The following project offers an innovative combination of the use of Generative AI in Next.js that can help in managing inventory logistics. This solution fills the void that exists between unstructured data and structured database management systems.

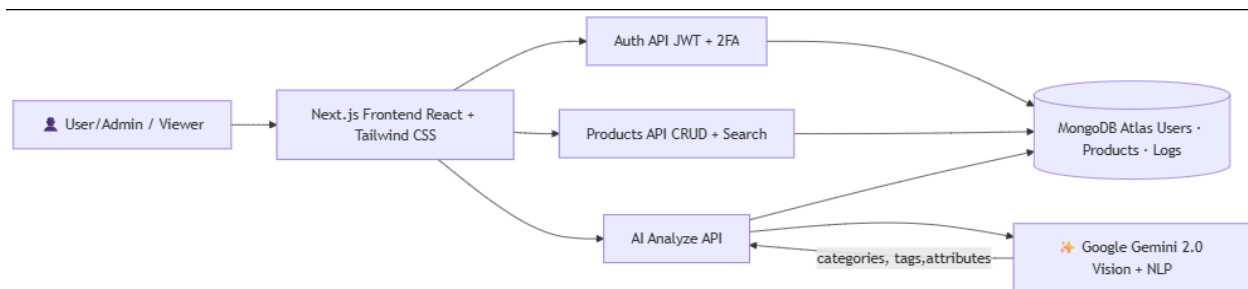
Technical Architecture

2.1 Tech Stack

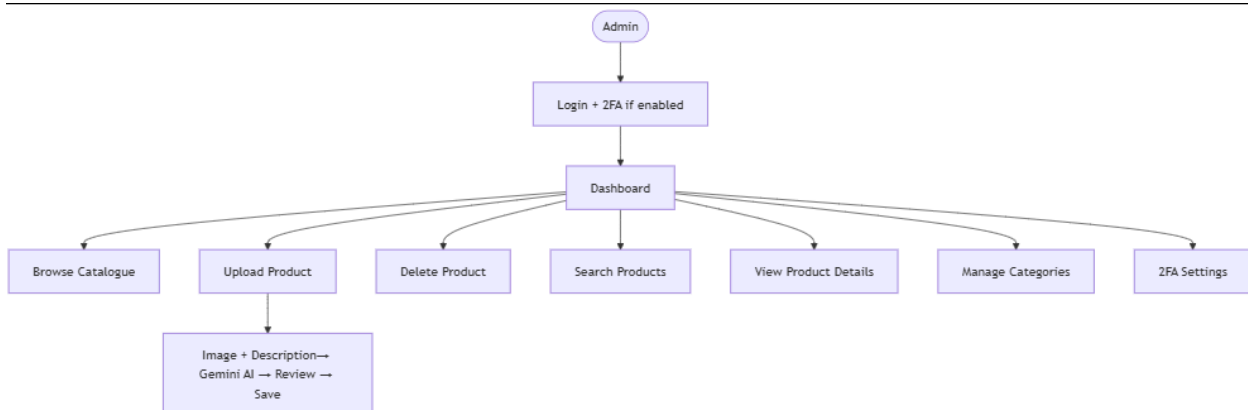
Layer/Component	Technology
Frontend	React 19, Tailwind CSS
Backend	Express API Routes
Database	Firebase db (cloud-hosted)
ORM	Firebase
Authentication	Firebase auth
2FA	TOTP via otpauth library
AI Engine	Google Gemini 2.0 Flash (@google/generative-ai)

Architectures and Diagrams

3.1 System Architectures



3.2 Admin flow Diagram



3.3 User/Viewer flow Diagram

